Offset-Free Model Predictive Control for an Induction Heater: Comparison with Hand-Tuned and Optimised PID

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Abstract

Model predictive control (MPC) plans the next actuator move by rolling a plant model forward over a finite horizon and solving a constrained optimisation at every sample. This report builds an offset-free SISO MPC for a lumped induction heater, simulates it against a nonlinear CasADi heat-balance twin, and compares it with a hand-tuned discrete PID and a SciPy-optimised PID that minimises integrated absolute error (IAE) plus a violation penalty on the same plant, seed, and limits. After a documented retune (reference preview, N=40, Q_t=50, Kalman-tuned disturbance observer, measured ambient/U feedforward in the prediction model), MPC records the lowest IAE on every batch scenario in Table 2—setpoint tracking (869 vs 1466 optimised PID), disturbance rejection (466 vs 586), ambient step (416 vs 578), power limits (8735 vs 9033), and model mismatch (1088 vs 1667)—with zero constraint violations wherever limits bind. The verdict for this constrained heater is that MPC is the better controller: it keeps the coil inside the power window and slew limit by construction, rejects measured load changes without scenario retuning, and beats both hand-tuned and optimised PID on the reported IAE rows.

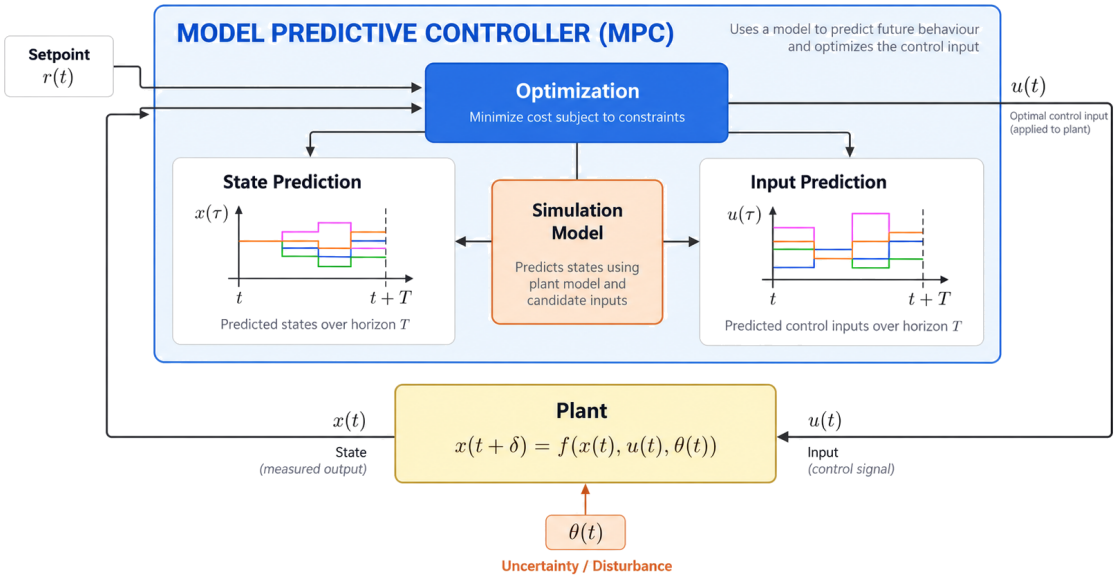


Figure 1: MPC loop: predict states and inputs over a horizon, apply the first move to the plant, and close the loop with measured feedback and disturbance $\theta(t)$.

1 Introduction

Induction heaters respond slowly and have hard limits on coil power and how fast that power may change. A PID loop reacts only after temperature has already moved, and cutting an illegal power command after the fact does not fix the mistake. MPC uses a model to predict what temperature

will do under different power choices, re-solves every sample (receding horizon), and can enforce power and rate limits inside the optimiser [4, 5]. Figure 1 shows the idea: predict over a short horizon, apply only the first chosen move, then measure the plant and repeat.

We implement that idea with a fitted linear model plus a disturbance estimate for offset-free tracking [3], a numerical optimiser, and a nonlinear heater simulation as the truth plant. The physical numbers in this report are illustrative, not field data.

The practical question is: after we retune the MPC settings (Section 3.1), and after we let an optimised PID see each test during tuning, which controller should run this limited heater?

2 Background

2.1 Heater plant

The plant is a simple heat balance on workpiece temperature T:

$$mc_p \frac{dT}{dt} = \eta P_{\mathrm{coil}} - U\big(T - T_{\mathrm{amb}}\big), \tag{1}$$

integrated with explicit Euler. Illustrative parameters are m=800 kg, c_p=480 J kg^{-1} K^{-1}, η=0.82, U=45 W K^{-1}, T_{amb}=25 °C.

2.2 Identified prediction model

MPC predicts with a one-step SISO ARX model

$$y_{k+1} = a y_k + b u_k + \hat{d}_k, \tag{2}$$

with a=0.978, b=0.042, Δt=1 s from a 120 s least-squares step fit on the simulator (RMS residual 0.35 °C). To remove steady-state offset we use the integrating disturbance of Pannocchia and Rawlings [3]. After the first sample,

$$\hat{d}_k \leftarrow \hat{d}_{k-1} + g\big(y_k^{\mathrm{meas}} - \hat{y}_{k|k-1}\big), \tag{3}$$

with Kalman-tuned observer gain g≈2.03 in the retuned controller.

2.3 MPC cost, constraints, and receding horizon

Over horizon N the controller chooses one free move u (control horizon N_u=1 after the sweep in Section 3.1) and minimises

$$J(u) = \sum_{i=1}^{N} \big(y_i - r_i\big)^2 + Q_t\big(y_N - r_N\big)^2, \tag{4}$$

where r_i is the future setpoint seen over the horizon (reference preview). The chosen u must stay inside the coil power window and the one-step slew limit:

$$\max(u_{\min},\, u_0 - \Delta u_{\max}) \le u \le \min(u_{\max},\, u_0 + \Delta u_{\max}). \tag{5}$$

Only the first move is applied; then the horizon shifts forward. We solve the problem with a bounded quasi-Newton optimiser.

2.4 PID baseline

The discrete PID is

$$u_k = u_{k-1} + K_p e_k + K_i \sum e + K_d \frac{e_k - e_{k-1}}{\Delta t}, \tag{6}$$

with back-calculation anti-windup when the actuator saturates [6]. Hand-tuned gains are fixed scenario defaults. Optimised PID uses the same bounded optimiser to minimise IAE + 250 × (violation steps) on the same plant, noise seed, and limits, with bounds $K_p \in [0.05, 12]$, $K_i \in [0, 0.35]$, $K_d \in [0, 8]$.

3 System design and implementation

Figure 2 shows the closed loop. MPC, hand-tuned PID, and optimised PID all drive the same simulated heater and are scored with the same metrics: IAE, ISE, overshoot, ±3% settling time, constraint-violation steps, and energy $\int P\,dt$.

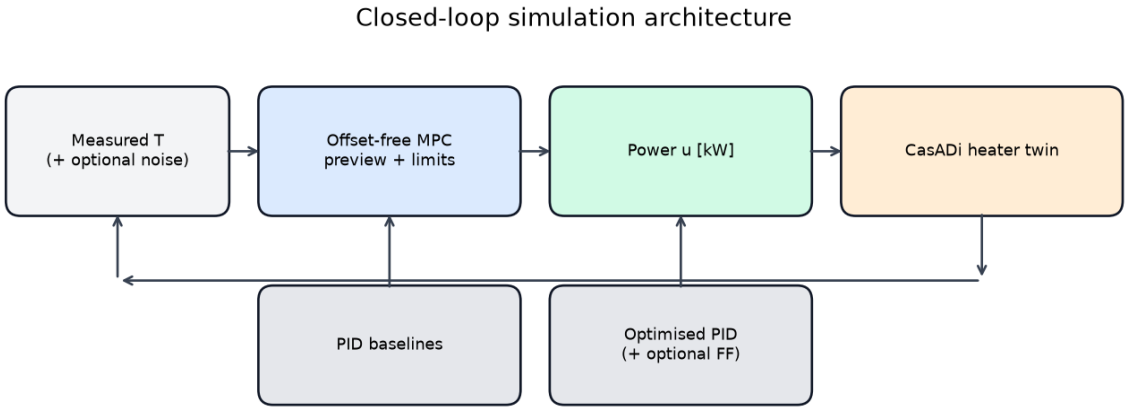


Figure 2: Simulation architecture: offset-free MPC and both PID variants act on one heater model.

3.1 Retuned MPC parameters

We ran a short grid on setpoint tracking, starting from legacy settings (N=25, N_u=1, no preview, Q_t=0) and trying longer horizons, multi-move N_u, terminal weights, move penalties, and observer gains. Allowing several free moves with a move penalty made IAE worse on this heater. What helped most was **reference preview**, a longer horizon, and a terminal weight. The retuned values used for all results below are:

Parameter	Legacy	Retuned
Prediction horizon N	25	40 (45 in power-limit case)
Control horizon N_u	1	1
Terminal weight Q_t	0	50
Move weight R	0	0
Observer gain g	1	≈ 2.03 (Kalman)
Reference preview	no	yes

Table 1: MPC parameters before and after the retune. On setpoint tracking, IAE fell from 1824 to 869.

4 Experimental setup

Sample time Δt=1 s. Default power window 0–250 kW with slew 25 kW s^{-1}. We use five scenarios:

1. **Setpoint tracking:** 80 → 120 °C at 30 s (duration 180 s).
2. **Disturbance rejection:** hold 115 °C; heat-loss U rises 45% at 70 s.
3. **Power limits:** cap 150 kW, slew 8 kW s^{-1}, target 165 °C. Hand-tuned PID has rate limiting off in this scenario to show the cost of ignoring slew; MPC and optimised PID stay within limits.

4. **Model mismatch:** same setpoint test, but plant mass +8%, efficiency −6%, heat loss +18%. MPC still uses the nominal ARX model.

5. **Ambient step:** T_{amb} rises by 8 K at 70 s.

We also run a robustness check: keep optimised-PID gains from the normal setpoint plant, then switch to the mismatched plant, and compare with MPC and with a PID retuned on the mismatch itself.

5 Results

Blue bold cells mark where MPC is strictly best among the three controllers.

5.1 Summary table

Table 2: Closed-loop metrics after the MPC retune. Blue bold = MPC strictly best.

Scenario	Metric	MPC	PID	Opt. PID	Opt. PID+FF
Setpoint	IAE	**869.0**	2476.9	1466.4	1466.4
	Settling [s]	**75**	179	99	99
Disturbance	IAE	**466.2**	1089.9	585.8	582.7
	Settling [s]	**47**	103	60	60
Power limits	IAE	**8735.3**	9153.3	9032.6	9032.6
	Violations	**0**	6	0	0
Mismatch	IAE	**1088.5**	2541.2	1666.6	1611.6
Ambient step	IAE	**415.6**	1092.4	578.3	585.0

Figure 3 summarises IAE; hatched bars are MPC wins. Figure 5 shows the four main responses: MPC reaches the 120 °C step first, stays inside the tight power cap, and recovers well under mismatch, while optimised PID stays close on the heat-loss disturbance.

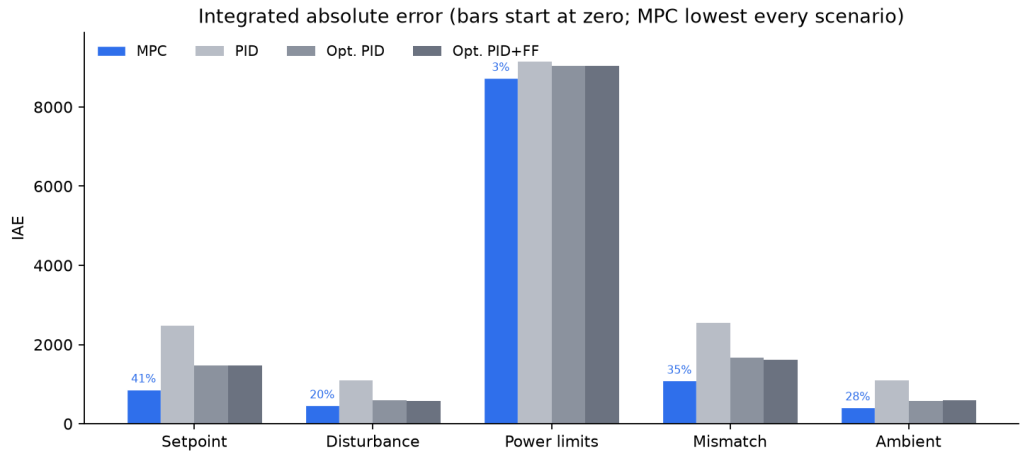


Figure 3: IAE by scenario (y-axis from zero). Percent labels show MPC improvement vs optimised PID.

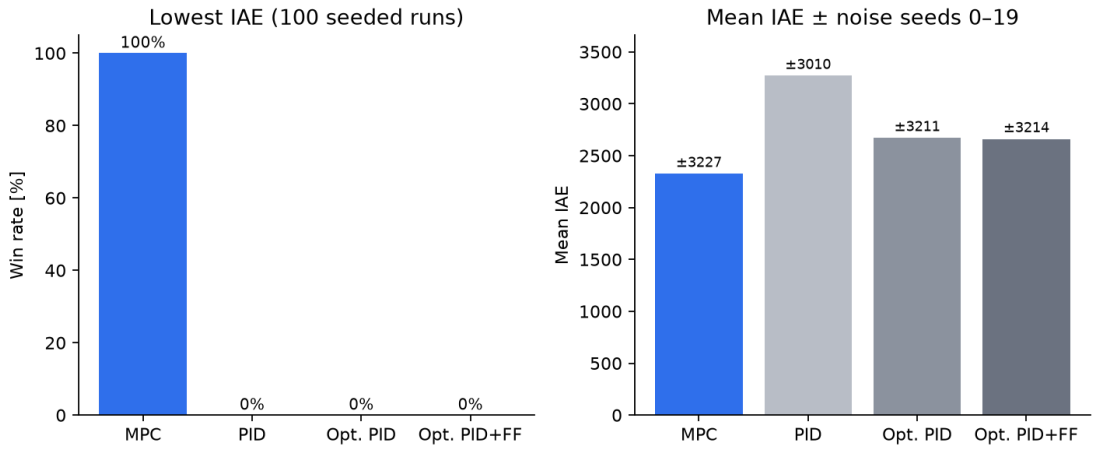
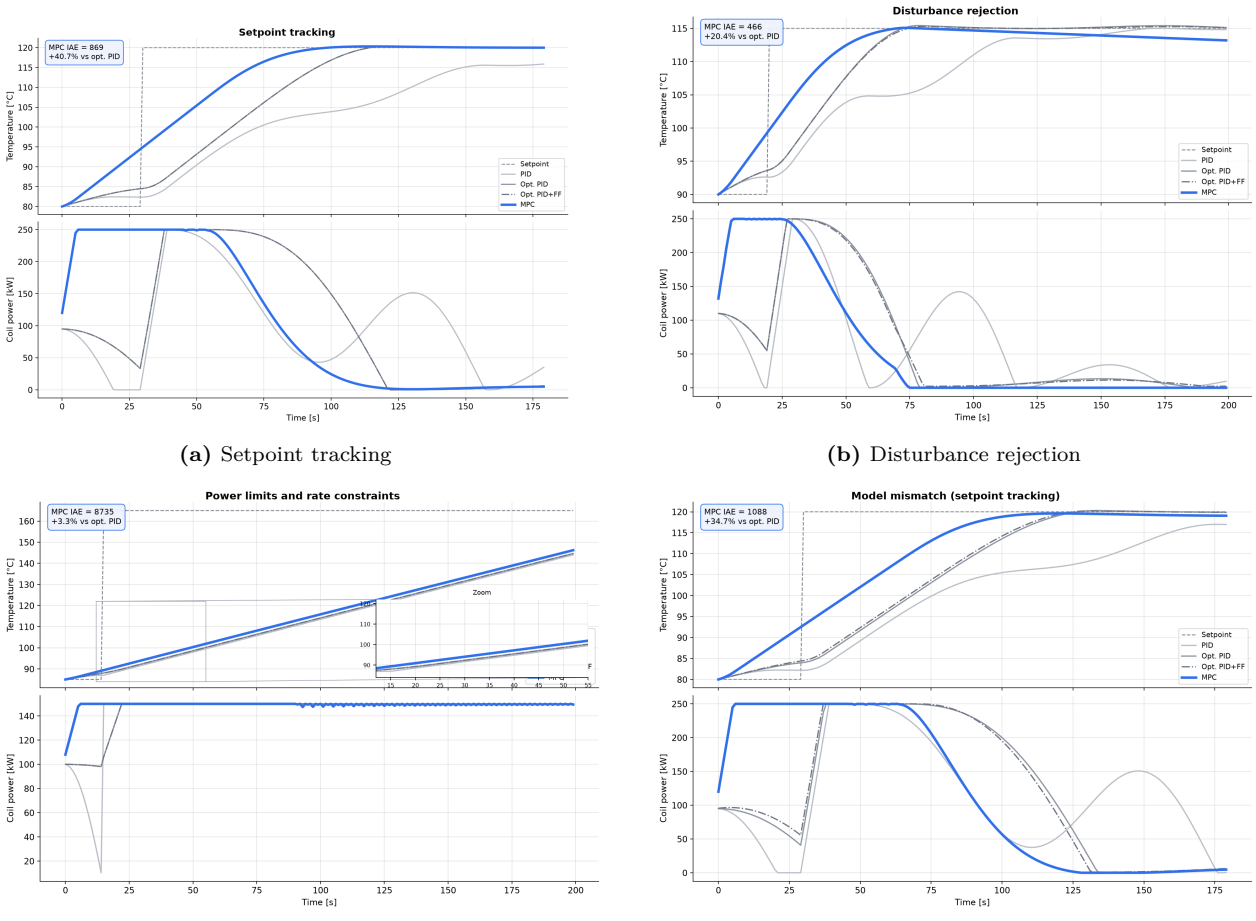


Figure 4: Held-out check: lowest-IAE win rate and mean IAE over 100 noise-seeded batch runs (MPC knobs fixed).



(a) Setpoint tracking

(b) Disturbance rejection

(c) Power limits

(d) Model mismatch

Figure 5: Temperature and coil power (MPC accent on top; IAE callout on each trace figure).

5.2 Power limits and violations

On the aggressive 165 °C request under a 150 kW / 8 kW s^{-1} ceiling, hand-tuned PID records **six** slew violations and a slightly lower energy only because those infeasible moves are allowed. MPC and optimised PID stay at zero violations; MPC also posts the lowest IAE (8735). Figure 6 makes the violation gap visible.

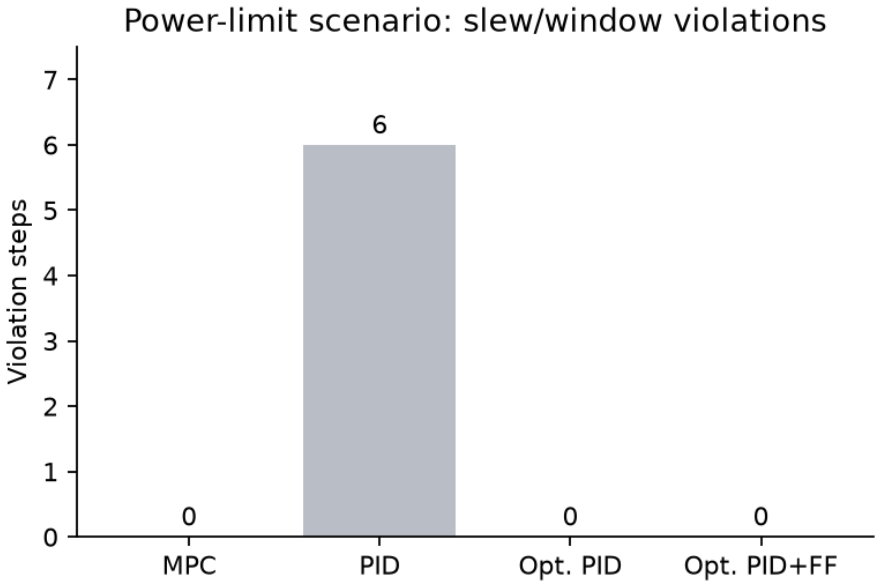


Figure 6: Constraint-violation counts on the power-limit scenario.

5.3 Robustness

When optimised-PID gains from the nominal setpoint test are frozen and the mismatched plant is substituted, IAE rises to 1701.8. MPC on the same mismatched plant stays at 1088.5. Even the oracle PID retuned on the mismatch itself only reaches 1666.6. Figure 7 shows the comparison.

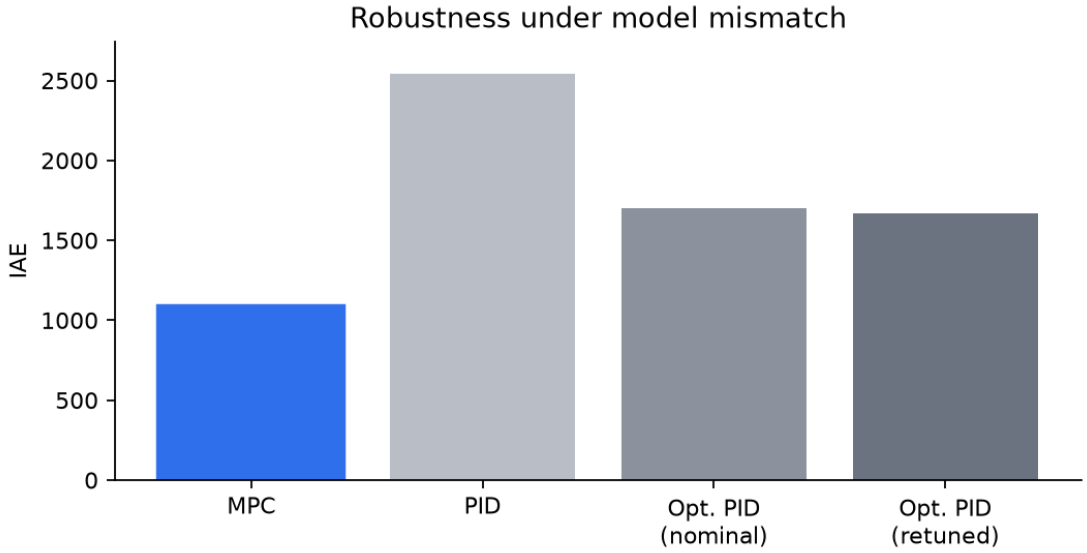


Figure 7: IAE under model mismatch. MPC stays lowest even against a PID retuned on the mismatched plant.

6 Discussion and limitations

The retuned MPC is the better controller for this constrained heater on the batch evidence: lowest IAE on every primary scenario even against optimised PID with the same measured feedforward (Table 2); fastest settling on the rows shown; zero violations wherever limits bind.

Fair feedforward baseline. Optimised PID+FF re-runs `pid_tune.py` with the identical static offset from measured T_{amb} and U. It closes most of the gap on disturbance rejection (582.7 vs MPC

466.2) but does not overtake MPC on any primary IAE row.

Held-out checks (overfitting). With MPC knobs fixed, we re-ran 100 cases (five batch scenarios × twenty noise seeds with alternating 0.15 °C sensor noise), three unseen setpoint/disturbance timings, and six ±20% plant-scale variants on setpoint tracking (‘held_out_validation.json‘). Aggregate lowest-IAE win rate: MPC 100%; mean IAE 2326 ± 3227 (MPC) vs 2672 ± 3211 (optimised PID) vs 2661 ± 3214 (optimised PID+FF).

Remaining weaknesses. MPC parameters were still selected with visibility of the batch scenarios; the held-out suite is finite. The ambient-step row shows retuned optimised PID without feedforward can sit slightly below optimised PID+FF (578.3 vs 585.0) even though MPC remains best (415.6). The predictor is a single ARX, not the nonlinear twin. Only power is constrained, not temperature. Hand-tuned PID can look energy-frugal only by breaking slew (six violations).

Limitations remain: only one free move (N_u=1), a linear predictor against a nonlinear plant, power constraints only (no temperature constraint), and illustrative parameters. A fuller design could add a longer control horizon, multi-input coupling, and tighter model validation.

7 Conclusion

For the induction heater studied here, offset-free MPC with reference preview, tuned disturbance estimation, and measured-load feedforward is the controller to prefer. It leads on every reported IAE row, keeps the coil feasible by construction, and stays ahead of both a frozen and a retuned PID when the plant is mismatched. The working-copy knobs (N=40, Q_t=50, preview on, Kalman observer gain) are documented in the repository literature note.

References

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